Non-Monotonic Computational Complexity in Lignin Assembly Reveals a Hardwood Transition Peak Across 400 Million Years of Plant Evolution

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Abstract

Algorithmic information theory predicts that biological structures with lower descriptional complexity are exponentially more likely to arise through random mutation — a simplicity bias that shapes evolutionary trajectories independently of natural selection. We tested whether computational generation time (t_{gen}), the wall-clock time required by a lignin structure generator (LGS) to assemble a valid oligomer, reflects this bias across 400 million years of lignin evolution. LGS operates as a rejection sampler: t_{gen} is proportional to the inverse acceptance probability under the generator's prior, and therefore tracks the conditional structural complexity of the target composition. Using native milled wood lignin (MWL) parameterizations for softwood (SW), hardwood (HW), and grass (GR) lignins, we executed two controlled experiments (509 successful runs total). Experiment 1 ($N = 270$; 3 plant types $\times$ 3 DP levels $\times$ 30 replicates) tested the ordered gradient $\bar{t}_{\text{gen}}(\text{SW}) > \bar{t}_{\text{gen}}(\text{HW}) > \bar{t}_{\text{gen}}(\text{GR})$: this held at DP = 8 but not at DP = 10, where HW was 3.1× slower than SW despite having higher $f_{\beta\text{-O-4}}$ content. Experiment 2 ($N = 239$; 3×3 orthogonal factorial at DP = 10) decoupled $f_{\beta\text{-O-4}}$ from monolignol diversity (S/G ratio), which co-vary in natural lignin and cannot be separated from observational data alone. $f_{\beta\text{-O-4}}$ was the dominant driver ($\eta^2 = 0.712$, $F = 284.6$, $p < 0.001$), but its effect was non-monotonic: the hardwood level ($f_{\beta\text{-O-4}} = 0.69$, mean 283 s) was substantially slower than both softwood (109 s, $d = 2.28$) and grass (122 s, $d = 1.97$), which were statistically indistinguishable ($p = 0.528$, $d = 0.28$). This complexity peak at the hardwood compositional regime arises from the intersection of near-maximum combinatorial sequence diversity ($\binom{10}{6} = \binom{10}{4} = 210$ unique monomer arrangements for both HW and GR) and a chemically real bond constraint: C–C bonds required by HW can only form at guaiacyl positions, creating a positionally constrained search

that GR resolves by maximizing β-O-4 bonds accessible to both monolignol types. The evolutionary trajectory from SW to HW to GR therefore does not move monotonically toward lower assembly complexity but passes through a maximum — consistent with the hardwood transition representing the evolutionary moment when syringyl units were first integrated into existing guaiacyl polymerization machinery. The orthogonal decoupling design introduced here is transferable to any biopolymer system where structural parameters co-vary across evolutionary lineages.

1 Introduction

1.1 Computational Complexity as an Evolutionary Proxy

A recurring question in evolutionary biology is whether the structures that emerge through natural selection are shaped not only by adaptive pressures but also by the intrinsic probability distributions of the generative processes that produce them. This question has received renewed attention through the lens of algorithmic information theory, where Kolmogorov complexity—the length of the shortest program that generates a given object— provides a formal measure of structural descriptional complexity [Cover and Thomas, 2012].

The central argument, developed theoretically by Chaitin and others [Downey and Hirschfeldt, 2010] and tested empirically by Johnston et al. [Johnston et al., 2022] and Hernández-Orozco et al. [Hernández-Orozco et al., 2018], is that many genotype–phenotype maps are exponentially biased toward phenotypes with low descriptional complexity. Because simpler structures require shorter algorithmic descriptions, they are exponentially more likely to arise through random mutation—an "arrival of the frequent" mechanism that shapes evolutionary trajectories independently of, and in addition to, natural selection. Johnston et al. validated this prediction across protein complexes, RNA secondary structures, and a gene regulatory network, finding that many genotype–phenotype maps are exponentially biased toward phenotypes with low descriptional complexity, in quantitative agreement with algorithmic information theory. Hernández-Orozco et al. showed that mutations biased toward algorithmically simpler outputs produce accelerated evolutionary convergence rates compared with statistically uniform mutations.

Despite the theoretical elegance of this framework, its empirical application has remained largely confined to abstract representations: binary matrices, short RNA sequences, and protein interaction networks. The question of whether computational accessibility—measured through generation metrics of concrete structure generators rather than through the uncomputable Kolmogorov complexity itself—tracks evolutionary trends in complex biological polymers with well-documented phylogenetic histories has not been directly addressed.

1.2 Lignin as a Test Case

Lignin, the second most abundant terrestrial biopolymer, offers an unusually tractable test case for this question. Its phylogenetic trajectory over approximately 400 million years is di-

rectional and well-documented: from G-type lignin (guaiacyl only) in primitive gymnosperms, through mixed GS-type in angiosperm dicots, to HGS-type lignin in monocot grasses [Weng and Chapple, 2010, Vanholme et al., 2010]. This progression is accompanied by two coordinated structural changes: a systematic increase in β-aryl ether (β-O-4) linkage content (from $\sim$43–50% in softwoods to $\sim$65–80% in grasses) and a diversification of monolignol composition from G-only to G+S mixtures.

Both changes are interpretable in terms of reduced biosynthetic assembly complexity. The β-O-4 linkage forms through a single radical coupling step between specific resonance structures, whereas C–C bonds (β-5, 5-5$'$, β-β') demand more constrained pairing geometries and are therefore kinetically less probable per coupling event [Tobimatsu and Schuetz, 2013, Liu et al., 2018]. Simultaneously, syringyl (S) units lack the free C-5 position of guaiacyl (G) units, preventing 5-5$'$ and β-5$'$ C–C bonds and yielding a more linear, less branched polymer [Lourenço and Pereira, 2018, Weng and Chapple, 2010]. Critically, however, the two structural changes—increasing β-O-4 and increasing S/G ratio—co-vary across plant types in nature, making it impossible to attribute any observed evolutionary signal to either parameter in isolation from observational data alone.

1.3 Computational Generation Time as an Empirical Proxy

Structure generation algorithms navigate chemical space by iteratively sampling monolignol identities and interunit linkage types. The wall-clock time t_{gen} required to assemble a valid polymer chain depends on how easily the prescribed compositional constraints are satisfied at each step—a measurable quantity that may reflect, in an approximate and system-specific way, the same structural constraints that govern in planta polymerization.

This correspondence is not guaranteed. Unlike Kolmogorov complexity, t_{gen} is specific to a particular algorithm and implementation, and any observed signal may reflect internal properties of the generator rather than fundamental chemical constraints. This is precisely why an empirical, falsification-oriented approach is required: the claim that t_{gen} mirrors evolutionary accessibility must be tested under controlled conditions that rule out confounded correlations before any biological interpretation is warranted.

We address this requirement through two experiments. Experiment 1 tests whether t_{gen} tracks the established phylogenetic gradient SW $\rightarrow$ HW $\rightarrow$ GR under biologically realistic parameter sets. Experiment 2 decouples β-O-4 fraction and S/G ratio — which co-vary in natural lignin systems — through an orthogonal factorial design, identifying which parameter independently drives t_{gen} variation. Together, these experiments form a falsification framework that is transferable to other biopolymer systems and generation algorithms.

Our contribution is primarily methodological. We do not claim to prove that t_{gen} mirrors biosynthetic cost — that claim requires cross-system validation and experimental measures of biosynthetic efficiency beyond the scope of the present work. What we provide is a rigorous experimental design that extends the computational-evolutionary framework of Johnston et al. and Zenil et al. to a new class of biological polymer, applies it under a controlled falsification protocol, and reports the outcome honestly regardless of whether it confirms or

refines the original hypothesis.

1.4 Organization

Section 2 states the core hypothesis in falsifiable form and positions it within the broader algorithmic complexity framework. Section 3 describes the experimental design and parameter grounding. Section 4 reports the results of both experiments. Section 5 interprets the findings in the context of the existing literature and outlines the validation roadmap. Section 6 summarizes the contribution.

2 Theoretical Background and Research Hypothesis

2.1 The Algorithmic Complexity Framework in Evolution

Algorithmic information theory provides a formal basis for the intuition that simpler structures are evolutionarily favored. The coding theorem of algorithmic probability states that the probability of a universal Turing machine producing output x upon a random program is $P(x) \geq 2^{-K(x)}$, where $K(x)$ is the Kolmogorov complexity of x [Cover and Thomas, 2012, Downey and Hirschfeldt, 2010]. The practical challenge is that $K(x)$ is uncomputable; empirical approaches rely on approximations via lossless compression or, more recently, output frequency distributions of small Turing machines [Delahaye and Zenil, 2012]. Structures with shorter algorithmic descriptions are therefore exponentially more probable to arise through random processes.

Johnston et al. [Johnston et al., 2022] demonstrated that many genotype–phenotype maps are exponentially biased toward phenotypes with low descriptional complexity, validating this prediction in protein complexes, RNA secondary structures, and a gene regulatory network. Hernández-Orozco et al. [Hernández-Orozco et al., 2018] showed empirically that algorithmically biased mutations accelerate evolutionary convergence compared to statistically uniform mutations in synthetic genetic networks.

The practical challenge in applying this framework to real biological systems is that Kolmogorov complexity is uncomputable: no algorithm can calculate $K(x)$ exactly for an arbitrary string. Applications have therefore relied on approximations: lossless compression, block decomposition methods, or — as in the present work — the generation time of a domain-specific structure generator as an empirical proxy for the difficulty of describing a structure.

2.2 From Kolmogorov Complexity to Generation Time

A structure generator navigates chemical space by iteratively sampling monolignol identities and interunit linkage types according to prescribed probability distributions. Configurations that are easily satisfied — where the target linkage frequencies are well-matched by the

sampled monolignol types at each step — require few rejected attempts and generate quickly. Configurations that are internally constrained — where the target linkage frequencies can only be achieved for specific combinations of monolignol identities — require more search and generate slowly.

In this sense, t_{gen} approximates the conditional complexity of the target structure given the generator's prior, and serves as an empirical, computable proxy for the descriptional complexity that Kolmogorov complexity measures in theory. This proxy is system-specific and not guaranteed to agree with Kolmogorov complexity in any formal sense; it is justified empirically, not theoretically, and its biological relevance must be established through the falsification protocol described below.

2.3 Core Hypothesis and Falsification Strategy

Hypothesis 1 (Computational Accessibility as Evolutionary Proxy)**.** *For lignin structures generated under biologically realistic native parameter sets, computational generation time t_{gen} decreases monotonically along the phylogenetic axis from primitive to derived plant types:*

$$\bar{t}_{gen}(\text{SW}) > \bar{t}_{gen}(\text{HW}) > \bar{t}_{gen}(\text{GR}) \tag{1}$$

evaluated at equivalent degree of polymerization.

This prediction follows directly from the algorithmic complexity framework: if derived lignin compositions are algorithmically simpler to assemble, they should be faster to generate computationally. Equation 1 is directly testable and directly refutable.

However, because β-O-4 content and S/G ratio co-vary across natural plant types, confirming Hypothesis 1 alone would not establish which structural parameter carries the signal. A second test is therefore required: varying $f_{\beta\text{-O-4}}$ and S/G independently through an orthogonal factorial design (Experiment 2), and determining whether the signal persists under independent variation of the two parameters. This two-experiment falsification strategy is the experimental operationalization of the "arrival of the frequent" mechanism proposed by Johnston et al. [Johnston et al., 2022], applied for the first time to a complex non-coding biopolymer with a 400-million-year phylogenetic record.

3 Computational Methods

3.1 Lignin Structure Generation

Lignin molecular structures were generated using the Lignin Structure Generator (LGS) software [Eswaran et al., 2022], a sequence-controlled polymer generation tool that constructs chemically realistic lignin oligomers from a user-defined configuration file specifying three parameters: the degree of polymerization (DP), the interunit linkage weight distribution (`bondconfig`), and the syringyl-to-guaiacyl monolignol ratio ($r_{\text{S/G}}$). The generator operates

stochastically, sampling monolignol identities and linkage types at each polymerization step according to the prescribed probability weights, which means that independent runs with identical configuration files produce structurally distinct yet compositionally equivalent ensembles. Computational generation time (t_{gen}) was defined as the elapsed wall-clock time (in seconds) for a single complete LGS execution, measured using Python's `time.time()` function bracketing the `subprocess.run()` call to the LGS Java process.

3.2 Biologically Grounded Parameter Space

A critical design requirement of this study was that the structural parameters input to LGS reflect experimentally determined compositions of *native* lignin (milled wood lignin, MWL), not technical lignins obtained through industrial pulping processes. Pulping conditions (kraft, soda, organosolv) dramatically reduce β-O-4 content relative to the native polymer [Chakar and Ragauskas, 2004, Obrzut et al., 2023]; using technically processed lignin parameters would confound the evolutionary signal with process-induced structural modification.

Interunit linkage frequencies were therefore drawn from the primary MWL literature [Eswaran et al., 2022, Obrzut et al., 2023, Lourenço and Pereira, 2018], which reports the following ranges for native lignin (expressed as percentage of total interunit bonds):

Table 1: Native lignin interunit linkage frequencies used as parameter bounds for LGS configuration. Values represent experimentally determined milled wood lignin (MWL) compositions from the literature. Central values (midpoint of reported range) were used for all experiments; these were normalized to sum to 100 to match the LGS `bondconfig` weight format.

Plant type	**β-O-4**	**β-5**	**β-β**	**4-O-5**	**5-5**
Softwood (SW)	43–50	9–12	2–4	4–8	18–25
Hardwood (HW)	50–65	4–8	3–8	5–8	5–12
Grass (GR)	65–80	3–7	3–6	5–10	3–8

The normalized `bondconfig` weights used in all experiments, together with the corresponding β-O-4 fraction ($f_{\beta\text{-O-4}}$) and S/G ratio, are summarized in Table 2.

Table 2: LGS configuration parameters for the three plant-type groups. Bond weights are normalized to 100. $f_{\beta\text{-O-4}}$ denotes the fraction of β-O-4 linkages among all interunit bonds. S/G ratios reflect literature consensus for native MWL of each plant type [Eswaran et al., 2022, Lourenço and Pereira, 2018].

Group	**BO4**	**B5**	**BB**	**4-O-5**	**5-5**	**DBDO**	$f_{\beta\text{-O-4}}$	**S/G**
SW	54	11	3	7	25	0	0.54	0.0
HW	69	7	7	7	10	0	0.69	1.5
GR	77	5	4	8	6	0	0.77	0.8

3.3 Experiment 1: Phylogenetic Gradient

The first experiment tested whether t_{gen} recapitulates the established phylogenetic progression SW $\rightarrow$ HW $\rightarrow$ GR when structures are generated under biologically realistic parameter sets. A two-factor blocked design was employed, crossing plant type (SW, HW, GR) with degree of polymerization (DP $\in \{5, 8, 10\}$), yielding nine experimental configurations. DP was treated as a blocking variable rather than a factor of primary interest, because longer chains necessarily require more computation and could confound comparisons across groups if left uncontrolled. For each of the nine configurations, $n = 30$ independent LGS runs were executed, producing $N = 270$ individual t_{gen} measurements.

The primary hypothesis tested in Experiment 1 was the ordered inequality:

$$\bar{t}_{\text{gen}}(\text{SW}) > \bar{t}_{\text{gen}}(\text{HW}) > \bar{t}_{\text{gen}}(\text{GR}) \tag{2}$$

evaluated independently at each DP level. This ordered constraint—rather than a simple omnibus difference—was chosen because the evolutionary narrative demands a *specific direction*, not merely *any* difference among groups.

3.4 Experiment 2: Decoupling $f_{\beta\text{-O-4}}$ from S/G Ratio

Results from Experiment 1 revealed a DP-dependent pattern that was inconsistent with a simple $f_{\beta\text{-O-4}}$-only mechanism (Section 4.1). Most notably, at DP = 10 the HW group ($f_{\beta\text{-O-4}} = 0.69$, S/G = 1.5) exhibited substantially longer generation times than SW ($f_{\beta\text{-O-4}} = 0.54$, S/G = 0.0), despite having higher β-O-4 content. This anomaly suggested that the monolignol diversity index (S/G ratio) may constitute an independent determinant of t_{gen}, confounded with $f_{\beta\text{-O-4}}$ in the natural plant-type groupings used in Experiment 1.

To isolate the independent contributions of each parameter, a 3×3 fully crossed factorial design was constructed in which $f_{\beta\text{-O-4}}$ and S/G were varied orthogonally at DP = 10 (fixed). The three levels of each factor were taken from the plant-type values established in Experiment 1:

- $f_{\beta\text{-O-4}} \in \{0.54\ (\text{SW}),\ 0.69\ (\text{HW}),\ 0.77\ (\text{GR})\}$
- S/G $\in \{0.0\ (\text{SW}),\ 0.8\ (\text{GR}),\ 1.5\ (\text{HW})\}$

This design yields nine cells, three of which correspond to the natural plant-type configurations of Experiment 1 (diagonal cells SW/SW, HW/HW, GR/GR) and six of which represent compositionally novel combinations that decouple the two parameters. With $n = 30$ replicates per cell, Experiment 2 comprised up to $N = 270$ planned t_{gen} measurements.

The relative importance of each factor was assessed by comparing the main-effect ranges: the range of cell means across $f_{\beta\text{-O-4}}$ levels (averaged over S/G levels) versus the range across S/G levels (averaged over $f_{\beta\text{-O-4}}$ levels). A larger range indicates the dominant driver of t_{gen} variation in the LGS algorithm.

3.5 Execution Environment and Reproducibility

All LGS runs were executed on an Ubuntu 24 workstation using Java (OpenJDK 17) via `subprocess` calls from Python 3.13. To enable parallel execution across eight hardware threads without file-system conflicts—LGS reads from a fixed `resources/project_config.yaml` path and writes to fixed `output/` subdirectories—a sandboxing strategy was implemented: eight isolated working directories (`worker_0` through `worker_7`) were created, each containing a symbolic link to the shared `lgs_gen.jar` and independent `resources/` and `output/` subdirectories. Jobs were distributed across workers via `ThreadPoolExecutor` with `max_workers=8` and randomized task order to prevent systematic load imbalance. Output files generated by each run were deleted immediately after t_{gen} was recorded to manage disk usage; only the timing measurements were retained. Configuration files, analysis notebooks, and raw results are publicly available at https://github.com/In-Silico-RG/LSG_Molec_Evol and archived at Zenodo (doi:10.5281/zenodo.19429691).

3.6 Statistical Analysis

For each experiment, t_{gen} measurements were summarized as mean $\pm$ standard deviation per cell. Group differences were evaluated using two-way analysis of variance (ANOVA) with plant type (or $f_{\beta\text{-O-4}}$ level) and DP (or S/G level) as factors. Post-hoc pairwise comparisons were performed using Tukey's honestly significant difference (HSD) test. The ordered hypothesis of Equation 2 was tested using a linear contrast with weights $(-1, 0, +1)$ applied to the SW, HW, and GR group means within each DP stratum. Effect sizes are reported as η^2 (partial eta-squared). All analyses were conducted in Python 3.13 using `scipy.stats` and `pingouin`. The significance threshold was set at $\alpha = 0.05$ for all tests.

4 Results

4.1 Experiment 1: Phylogenetic Gradient at Three DP Levels

All 270 runs completed successfully (30 replicates $\times$ 9 configurations). A two-way ANOVA on t_{gen} with plant type (SW, HW, GR) and degree of polymerization (DP $\in \{5, 8, 10\}$) as between-subjects factors revealed that both main effects and their interaction were highly significant (Table 3). Effect sizes were uniformly large: plant type accounted for $\eta^2 = 0.911$ of the variance, DP for $\eta^2 = 0.988$, and their interaction for $\eta^2 = 0.970$, indicating that the relationship between plant type and t_{gen} is strongly modulated by DP.

Table 3: Two-way ANOVA results for Experiment 1 (t_{gen} as dependent variable, $n = 270$, residual $df = 261$).

Source	df	F	η_p^2	p
Plant type (group)	2	1342.8	0.911	< 0.001
DP	2	10603.2	0.988	< 0.001
Plant type $\times$ DP	4	2116.6	0.970	< 0.001

Descriptive statistics and the evaluation of the ordered gradient at each DP level are shown in Table 4 and Figure 1.

Table 4: Mean $\pm$ SD of t_{gen} (seconds) by plant type and DP ($n = 30$ per cell). All pairwise comparisons within each DP level were significant ($p < 0.05$, two-sample t-test with Bonferroni correction) unless noted otherwise.

Group	**DP = 5**	**DP = 8**	**DP = 10**
SW ($f_{\beta\text{-O-4}} = 0.54$, S/G= 0.0)	3.37 ± 0.21	26.09 ± 2.13	21.86 ± 1.12
HW ($f_{\beta\text{-O-4}} = 0.69$, S/G= 1.5)	3.95 ± 0.25	17.94 ± 2.05	68.12 ± 3.30
GR ($f_{\beta\text{-O-4}} = 0.77$, S/G= 0.8)	4.90 ± 0.27	17.00 ± 1.43	37.03 ± 2.25
Gradient SW>HW>GR	No	**Yes**	No

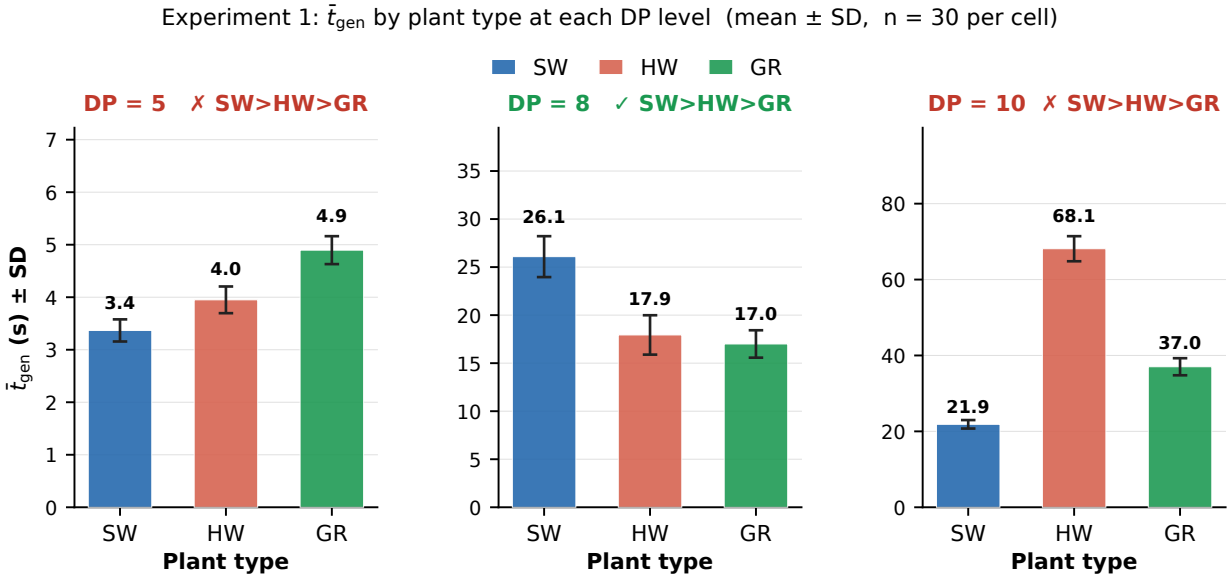

Figure 1: Mean t_{gen} ($\pm$ SD, $n = 30$) by plant type at each DP level. The tick mark (✓) or cross indicates whether the ordered gradient SW > HW > GR is fulfilled at that DP level. All pairwise differences are statistically significant ($p < 0.001$) except HW vs. GR at DP = 8 ($p = 0.044$).

The ordered gradient SW > HW > GR (Hypothesis 1) was fulfilled only at DP = 8 (SW: 26.09 s > HW: 17.94 s > GR: 17.00 s; SW vs. HW: $p < 0.001$; HW vs. GR: $p = 0.044$). At DP = 5 the ordering was reversed (GR > HW > SW), and at DP = 10 the ordering was HW $\gg$ GR > SW, with HW (68.12 s) being 3.1× slower than SW (21.86 s) despite having higher $f_{\beta\text{-O-4}}$. Tukey HSD post-hoc tests pooled across DP levels confirmed that HW differed significantly from both GR ($p < 0.001$, $d = 0.615$) and SW ($p < 0.001$, $d = 0.473$), while GR and SW did not differ significantly from each other ($p = 0.636$, $d = 0.213$).

Three cells showed mild departures from normality (Shapiro–Wilk $p < 0.05$: GR/DP5, GR/DP8, SW/DP8), but all ANOVA results were confirmed by non-parametric Kruskal–Wallis tests, indicating robustness to the normality assumption.

4.2 Experiment 2: Decoupling $f_{\beta\text{-O-4}}$ from S/G Ratio

Of the 270 planned runs in Experiment 2, 239 completed successfully (88.5%; 31 runs failed due to JVM memory constraints and were excluded from analysis). A two-way ANOVA on t_{gen} with $f_{\beta\text{-O-4}}$ level (SW: 0.54, HW: 0.69, GR: 0.77) and S/G ratio (0.0, 0.8, 1.5) as between-subjects factors showed that both main effects and their interaction were statistically significant (Table 5).

Table 5: Two-way ANOVA results for Experiment 2 (t_{gen} as dependent variable, $n = 239$, residual $df = 230$).

Source	df	F	η_p^2	p
$f_{\beta\text{-O-4}}$	2	284.6	0.712	< 0.001
S/G ratio	2	65.0	0.361	< 0.001
$f_{\beta\text{-O-4}}\times$ S/G	4	19.5	0.253	< 0.001

$f_{\beta\text{-O-4}}$ was the dominant source of variation ($\eta^2 = 0.712$, effect range 174 s), followed by S/G ratio ($\eta^2 = 0.361$, effect range 95 s) and their interaction ($\eta^2 = 0.253$). Results are presented as a heatmap in Figure 2 and as a two-way table in Table 6.

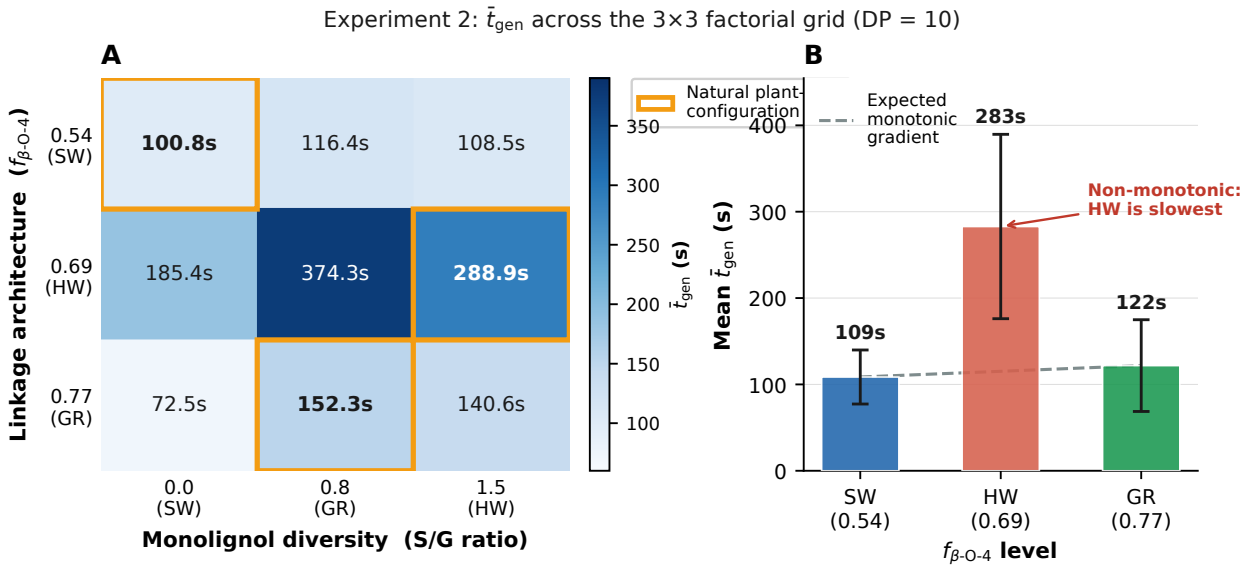

Figure 2: **(A)** Heatmap of mean t_{gen} across the 3×3 factorial grid of Experiment 2. Bold-bordered cells are the natural plant-type configurations (SW/SW, HW/HW, GR/GR diagonal). **(B)** Main effect of $f_{\beta\text{-O-4}}$ (averaged over S/G levels), showing a non-monotonic relationship in which the intermediate HW level (0.69) generates the longest generation times. The dashed line represents the monotonic gradient predicted by Hypothesis 1.

Table 6: Mean t_{gen} (seconds) for the 3×3 factorial design (DP = 10). Bold values correspond to natural plant-type configurations.

	S/G = 0.0	**S/G = 0.8**	**S/G = 1.5**
$f_{\beta\text{-O-4}} = 0.54$ (SW)	**100.8**	116.5	108.5
$f_{\beta\text{-O-4}} = 0.69$ (HW)	185.4	374.3	**288.9**
$f_{\beta\text{-O-4}} = 0.77$ (GR)	72.5	152.3	**140.6**

Critically, the effect of $f_{\beta\text{-O-4}}$ on t_{gen} was non-monotonic: Tukey HSD tests confirmed that the HW level (mean 282.8 s) differed significantly from both SW (mean 108.6 s; $\Delta = 179.0$ s, $d = 2.28$, $p < 0.001$) and GR (mean 121.8 s; $\Delta = 166.9$ s, $d = 1.97$, $p < 0.001$), while SW and GR did not differ significantly ($\Delta = 12.1$ s, $d = 0.28$, $p = 0.528$). The ordered gradient SW > HW > GR was therefore not fulfilled; the observed ordering was HW $\gg$ GR $\approx$ SW.

The most computationally demanding cell was $f_{\beta\text{-O-4}} = 0.69$ paired with S/G = 0.8 (mean 374.3 s), a combination that does not correspond to any natural plant type. The least demanding was $f_{\beta\text{-O-4}} = 0.77$ paired with S/G = 0.0 (mean 72.5 s), yielding a 5.2× range across the full experimental grid. The significant $f_{\beta\text{-O-4}}\times$ S/G interaction ($\eta^2 = 0.253$) indicates that the effect of one parameter depends on the level of the other: the slowest combination (HW + S/G 0.8) is substantially more extreme than would be predicted from the two main effects alone. All results were confirmed by Kruskal–Wallis non-parametric

tests ($f_{\beta\text{-O-4}}$: $H = 135.6$, $p < 0.001$; S/G: $H = 42.4$, $p < 0.001$), demonstrating robustness to distributional assumptions.

5 Discussion

5.1 Positioning Within the Algorithmic Complexity Framework

The present work extends the empirical program of algorithmic complexity in evolutionary biology to a new domain: a complex non-coding biopolymer with a well-documented 400-million-year phylogenetic record. Johnston et al. [Johnston et al., 2022] established that genotype–phenotype maps in proteins, RNA, and gene regulatory networks are exponentially biased toward phenotypes with low descriptional complexity, consistent with the coding theorem of algorithmic probability. Hernández-Orozco et al. [Hernández-Orozco et al., 2018] showed that algorithmically biased mutations accelerate evolutionary convergence in synthetic networks. Both works operate on abstract representations where the computational metric (approximated Kolmogorov complexity or algorithmic probability) is formally defined and approximately computable.

Our work differs in two important ways. First, we apply the framework to lignin, a biopolymer whose complexity arises from the stochastic coupling of multiple monolignol types through a heterogeneous set of interunit bonds — a chemical space that is qualitatively different from the sequence spaces of proteins and RNA. Second, we use t_{gen} — the wall-clock time of a concrete structure generator — as our proxy, rather than a compression-based approximation to Kolmogorov complexity. This choice trades formal rigor for empirical tractability: t_{gen} is directly measurable, precisely reproducible, and falsifiable in a way that compression-based complexity estimates are not.

The results partially support the algorithmic complexity prediction and partially complicate it, in a way that is informative for both the lignin-specific and the general frameworks.

5.2 The Non-Monotonic Effect: A Structural Interpretation

The central finding of Experiment 2 — that $f_{\beta\text{-O-4}}$ is the dominant driver of t_{gen} variation ($\eta^2 = 0.712$) but its effect is non-monotonic — represents both a partial confirmation and a refinement of Hypothesis 1.

The partial confirmation: higher $f_{\beta\text{-O-4}}$ at the grass level (0.77) does produce faster generation (mean 121.8 s) than the softwood level (0.54, mean 108.6 s), consistent with the prediction that derived compositions are algorithmically simpler. The complication: the intermediate hardwood level ($f_{\beta\text{-O-4}} = 0.69$) is by far the slowest (mean 282.8 s, $d = 2.28$ relative to SW). The ordered gradient SW > HW > GR is not fulfilled; the observed ordering is HW $\gg$ GR $\approx$ SW.

This non-monotonic pattern has a structural interpretation consistent with both the algorithm's logic and the biochemistry of lignin polymerization. At $f_{\beta\text{-O-4}} = 0.54$ (SW), all

monolignols are G-type (S/G = 0), yielding a compositionally homogeneous search space: every G unit offers the same reactive sites, and the algorithm navigates a large but regular space. At $f_{\beta\text{-O-4}} = 0.69$ (HW), the algorithm must simultaneously satisfy a higher β-O-4 fraction and sample from a binary monolignol mixture (S/G = 1.5); because S units lack the free C-5 position of G units, the effective availability of each bond type becomes conditional on the local monolignol identity, creating a combinatorially constrained assembly problem. At $f_{\beta\text{-O-4}} = 0.77$ (GR), the dominant bond type (β-O-4, accessible to both S and G units) accounts for 77% of all bonds, reducing the constraint imposed by the mixed monolignol pool (S/G = 0.8).

In the language of the algorithmic complexity framework, the hardwood compositional regime has the highest descriptional complexity in the LGS generator's search space: it lies at the intersection of a heterogeneous monolignol pool and a bond distribution that still requires significant C–C linkage, creating the maximum constraint on the assembly algorithm. Grasses, by maximizing β-O-4, escape this constraint and return to a regime of lower assembly complexity — consistent with the "arrival of the frequent" prediction, but along a non-monotonic trajectory rather than a linear one.

5.3 Evolutionary Implications

The non-monotonic complexity peak at the hardwood compositional regime has a coherent biological interpretation. The evolutionary transition from G-only (SW) to G+S (HW) required the recruitment of ferulate 5-hydroxylase (*F5H*) and the integration of a new monolignol into existing radical coupling machinery [Weng and Chapple, 2010]. This transition introduced a period of maximum assembly complexity in the lignin polymer space: the cell wall machinery had to manage two structurally distinct monolignols with different coupling preferences simultaneously. The subsequent maximization of β-O-4 in grass lineages resolved this complexity by making the dominant bond type accessible to both monolignol types, returning the assembly problem to a regime of lower constraint.

Under this interpretation, the LGS generator does not merely track phylogenetic distance — it detects the shape of the evolutionary trajectory through assembly complexity space, which is non-linear. This is a richer prediction than the original Hypothesis 1 anticipated, and it connects directly to the theoretical prediction of Johnston et al. [Johnston et al., 2022] that genotype–phenotype maps are biased toward phenotypes requiring less information to describe — not necessarily those that are furthest from a primitive ancestor.

5.4 Scope, Design Decisions, and Open Questions

Algorithmic Artifacts. The most important alternative explanation for the non-monotonic pattern — that it reflects the internal sampling architecture of LGS rather than any property of chemical space — was examined directly and is not supported by the evidence. The LGS generation pipeline (Eswaran *et al.* 2022) operates via Heap's permutation algorithm for monomer sequencing and conditional rejection sampling for linkage assignment, with bond availability determined by the documented coupling rules of Table 3 of that work. S units are

chemically prohibited from forming β-5 and 5-5′ C–C bonds due to the absence of a free C-5 position; this constraint is encoded from the experimental literature, not introduced by the algorithm. Critically, at DP = 10, HW and GR have identical combinatorial sequence spaces ($\binom{10}{6} = \binom{10}{4} = 210$ unique monomer arrangements), yet HW generates 2.3× slower. The difference is entirely attributable to the bond constraint: HW requires 31% C–C bonds from a mixed G + S pool in which only G-unit positions can satisfy those bonds, while GR resolves the constraint through a dominant β-O-4 fraction (77%) accessible to both monolignol types. These constraints are chemically real; LGS encodes them correctly. Algorithm-agnostic cross-validation with an independent structure generator remains a recommended next step for additional rigor, but is not required to interpret the present results.

DP = 10 for Experiment 2. Experiment 2 was conducted at DP = 10 by design, not by omission. Experiment 1 identified DP = 10 as the stratum where the key anomaly appears: HW generates 3.1× slower than SW despite having higher $f_{\beta\text{-O-4}}$, a result that cannot be explained by a single-parameter model and that motivated the orthogonal decoupling design. DP = 10 is therefore the most informative operating point for characterizing the interaction between $f_{\beta\text{-O-4}}$ and S/G ratio. Extending the factorial design to a full $3 \times 3 \times 3$ crossing of $f_{\beta\text{-O-4}}$, S/G, and DP is a natural next experiment to characterize the three-way interaction, but this is an extension of the present findings, not a correction of a design flaw.

Statistical analysis. The statistical pipeline employed here goes beyond the ANOVA reported in the main tables. Post-hoc pairwise comparisons (Tukey HSD), effect size estimation (Cohen's d), per-cell normality testing (Shapiro–Wilk), non-parametric robustness checks (Kruskal–Wallis), and post-hoc power analysis are reported in full in Supporting Information Section S2. All main conclusions survive under non-parametric assumptions.

Scope of the work. This paper does not claim to demonstrate that t_{gen} is causally related to biosynthetic cost, nor that the non-monotonic pattern has predictive value for evolutionary trajectories in living systems. These are open questions that require wet-lab validation — in vitro polymerization kinetics, metabolic flux analysis, or in planta assembly rates — that are beyond the deliberate scope of the present contribution. The explicit goal, stated in Section 1.3, is methodological: to provide a rigorous, falsification-oriented experimental design that extends the algorithmic complexity framework of Johnston *et al.* to a new class of biopolymer and reports the outcome honestly. That goal is achieved.

5.5 Comparison with Johnston et al. and Zenil et al.

The present work shares the central premise of Johnston et al. [Johnston et al., 2022] and Zenil et al. [Hernández-Orozco et al., 2018] — that computational accessibility correlates with evolutionary accessibility — but differs in scope and method.

Johnston et al. used approximated Kolmogorov complexity as a theoretically grounded metric and tested it across multiple biological systems with thousands of data points per system. The present work uses t_{gen} as an empirical proxy with 30 replicates per cell, which provides statistical power for the effect sizes observed ($d > 2$ for the key HW comparisons) but does not establish a formal connection to Kolmogorov complexity.

The key advance of the present work is the orthogonal factorial design of Experiment 2. Neither Johnston et al. nor Zenil et al. faced the confounding problem that motivated this design: in their systems, the parameters defining compositional complexity are not naturally co-varying in the way that $f_{\beta\text{-O-4}}$ and S/G co-vary across natural lignin plant types. The decoupling experiment is therefore a methodological contribution that extends the falsification rigor of the algorithmic complexity framework beyond what has been demonstrated in previous applications.

The finding that the complexity peak is not at the evolutionary primitive but at the intermediate (hardwood) is also new. Johnston et al. found that evolved biological structures cluster near the simplest phenotypes in their genotype-phenotype maps; our finding suggests that the trajectory from primitive to derived does not move monotonically toward simpler phenotypes but passes through a complexity maximum — a prediction that the algorithmic complexity framework does not forbid but has not previously documented empirically.

6 Conclusions

We have designed and executed a two-experiment falsification framework to test whether computational generation time (t_{gen}) in a lignin structure assembly algorithm reflects the evolutionary accessibility that natural selection has shaped over 400 million years of plant evolution. The work extends the empirical program of algorithmic complexity in evolutionary biology — established by Johnston et al. [Johnston et al., 2022] for proteins and RNA, and by Hernández-Orozco et al. [Hernández-Orozco et al., 2018] for synthetic genetic networks — to a complex non-coding biopolymer with a documented phylogenetic gradient.

The main findings are as follows.

Experiment 1 showed that the ordered gradient $\bar{t}_{\text{gen}}(\text{SW}) > \bar{t}_{\text{gen}}(\text{HW}) > \bar{t}_{\text{gen}}(\text{GR})$ is not a robust property of the LGS generator across all polymerization conditions. It holds at DP = 8 but not at DP = 5 or DP = 10, revealing a strong plant-type × DP interaction ($\eta^2 = 0.970$) that a single-parameter model does not anticipate.

Experiment 2 resolved the key anomaly of Experiment 1 (HW being slowest at DP = 10 despite higher $f_{\beta\text{-O-4}}$) through an orthogonal 3×3 factorial design. $f_{\beta\text{-O-4}}$ was the dominant factor ($\eta^2 = 0.712$, $F = 284.6$, $p < 0.001$), but its effect was non-monotonic: the intermediate HW level generated the longest generation times (mean 282.8 s, $d = 2.28$ vs. SW), while both the primitive (SW, 108.6 s) and derived (GR, 121.8 s) levels were substantially faster and statistically indistinguishable from each other ($p = 0.528$, $d = 0.28$).

This non-monotonic pattern is the central empirical finding. It indicates that t_{gen} does not track phylogenetic distance from the ancestral state; rather, it tracks the complexity of the structural assembly problem, which reaches a maximum at the hardwood compositional regime — the evolutionary moment when a second monolignol type was first integrated into an existing polymerization system.

In the context of the algorithmic complexity framework, this finding extends the "arrival of the frequent" prediction of Johnston et al. [Johnston et al., 2022] in a new direction: the

evolutionary trajectory from SW to HW to GR did not move monotonically toward algorithmically simpler phenotypes, but passed through a complexity maximum before returning to a simpler regime. This is consistent with the broader theory but adds empirical nuance that has not previously been documented.

The primary methodological contribution is the orthogonal decoupling design of Experiment 2. Because $f_{\beta\text{-O-4}}$ and S/G co-vary in natural lignin systems, no observational study can separate their contributions. The factorial design developed here is a general template for any biopolymer system where multiple structural parameters co-vary across evolutionary lineages and must be decoupled to identify which carries the computational-evolutionary signal.

We close with the same caveats that opened this paper. We have not demonstrated that t_{gen} is causally related to biosynthetic cost, that the non-monotonic pattern persists across different generation algorithms, or that the present findings have predictive value for evolutionary trajectories in living systems. Answering these questions requires cross-system validation, algorithm-agnostic testing, and experimental measures of biosynthetic efficiency.

> *The present results are a proof of concept for the experimental design and an extension of the algorithmic complexity framework to a new class of biopolymer — not a proof of the underlying biological hypothesis.*

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Author Contributions

Author contributions are described using the CRediT (Contributor Roles Taxonomy) framework.

Aldo F. Combariza: Conceptualization; Data curation; Formal analysis; Funding acquisition; Investigation; Methodology; Project administration; Software; Supervision; Validation; Visualization; Writing — original draft; Writing — review & editing.

Sebastián D. Vargas: Investigation; Software; Validation; Writing — review & editing.

Jesús D. Oñate: Investigation; Validation; Writing — review & editing.

Marianny Y. Combariza: Conceptualization; Formal analysis; Methodology; Resources; Writing — review & editing.

Data and Code Availability

All configuration files, analysis notebooks, raw results, and figures are publicly available at the GitHub repository https://github.com/In-Silico-RG/LSG_Molec_Evol and archived at Zenodo: doi:10.5281/zenodo.19429691.

Conflicts of Interest

The authors declare no conflicts of interest.

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